Supplementary Information for Fixed-Budget Local-Patch Risk Routing for Industrial Anomaly Detection

Jin Huang[1*] and Qisen Gao[2]

[1*]Guangxi Vocational Normal University, Nanning, 530007, Guangxi Zhuang Autonomous Region, China.
[2]Gansu University of Political Science and Law, Lanzhou, 730070, Gansu, China.

*Corresponding author(s). E-mail(s): 614938561@qq.com; ORCID: 0009-0005-7489-2018;
Contributing authors: 1350728839@qq.com;

1 Purpose and scope

This supplement provides details about the evaluation units, paired resampling, controlled baselines, failed variants, online routing, and latency measurements. Unless noted otherwise, the reported numbers come from the strict comparison with matched-random selection.

2 Evaluation units and uncertainty

Each evaluation unit pairs one dataset category with one random seed. The primary matched comparisons use 45 units for MVTec AD (15 categories × 3 seeds), 18 units for MPDD (6 × 3), and 36 units for VisA (12 × 3). Within each unit, local routing and matched-random selection use the same feature extractor, candidate patches, budget, seed, and downstream scoring rule. Reported effects are local-routing values minus matched-random values. Confidence intervals are percentile intervals from 10,000 paired bootstrap resamples of the evaluation units. The sign-count column gives the number of units with a positive effect.

Table 1 Primary paired effects at the 25% fallback budget

Dataset	n	AUROC Δ	95% CI	+ units	Recall Δ	95% CI
MVTec AD	45	0.1249	[0.0837, 0.1659]	33/45	0.3877	[0.2916, 0.4792]
MPDD	18	0.1166	[0.0715, 0.1612]	16/18	0.1975	[0.1097, 0.2939]
VisA	36	0.0128	[0.0081, 0.0171]	28/36	0.2300	[0.1336, 0.3214]

Pooling all 99 category and seed units gives an AUROC difference of +0.0826 (95% CI [0.0598, 0.1059]) and a Recall@5%FPR difference of +0.2958 ([0.2347, 0.3551]); 77 of the 99 units have a positive AUROC effect. Local-patch risk also exceeds the nearest-patch-distance dispersion controller by +0.0704 AUROC ([0.0446, 0.0990]). Its difference from the fast-score controller is +0.0017 (95% CI [-0.0184, 0.0233]), and fast-score is stronger on MPDD and VisA.

3 Budget sensitivity

The MVTec budget analysis uses the same three seeds and strict quota at 10%, 25%, and 50% fallback budgets, with 45 units at each budget. Table 2 reports paired AUROC and Recall@5%FPR differences from matched-random selection. The AUROC difference increases with the budget, whereas the Recall difference decreases; both remain positive at all three budgets. These results describe allocation quality relative to matched-random routing, not an end-to-end speedup.

Table 2 MVTec budget sensitivity (local-patch risk minus matched-random) at 10%, 25% and 50% fallback budgets

Budget	n	AUROC Δ	95% CI	Recall Δ	95% CI	+ units (AUROC / Recall)
10%	45	0.1145	[0.0821, 0.1484]	0.4981	[0.3940, 0.5977]	39/45, 42/45
25%	45	0.1249	[0.0837, 0.1659]	0.3877	[0.2916, 0.4792]	33/45, 42/45
50%	45	0.1589	[0.1226, 0.1954]	0.2479	[0.1834, 0.3110]	39/45, 35/45

4 Baseline and absolute-score context

Across the 45 MVTec units, the full detector obtains AUROC 0.9392. At the primary 25% local budget, the strict local route obtains 0.8343 and matched-random routing obtains 0.7094. These values place the routing result in context: the comparison concerns allocation under a reduced budget, not performance above the full detector. The PaDiM-style baseline obtains AUROC 0.8982 and Recall@5%FPR 0.6544. The controlled PatchCore-like patch-level farthest-point coreset detector obtains AUROC 0.9433 and Recall@5%FPR 0.7847. This implementation is a controlled approximation and is not intended to reproduce the official PatchCore release exactly.

5 Failure cases and boundary conditions

We retained three negative checks. A multi-scale rank-fusion variant underperformed matched-random routing in a three-category smoke test, showing that extra ranking signals can misallocate the budget. A lightweight cost filter underperformed the stronger patch-level coreset baseline. The conformal-threshold diagnostic produced a 54.9% fallback rate for a nominal 25% target, so it cannot serve as a budget guarantee. These checks limit the claim to improved selection under the fixed experimental budget; they do not establish that every controller or online policy is robust.

6 Online-prefix sensitivity

We evaluated the online prefix controller separately because its temporal prefix and fallback rule differ from the strict offline quota. Across the 45 MVTec units, the realized fallback rate was 25.34%, with an AUROC effect of 0.0331 (95% CI [0.0045, 0.0601]) and a Recall effect of 0.0804 (95% CI [0.0211, 0.1346]). On seed 17, the AUROC/Recall effects were 0.0890/0.0643 for MPDD and 0.00004/-0.0408 for VisA. With a VisA buffer of 128, the effects were 0.00124 (95% CI [-0.00113, 0.00339]) and 0.0750 (95% CI [0.0189, 0.1317]). The outcome therefore depends on the controller configuration and does not show a universal online improvement.

7 Latency measurement

Latency was measured on the same machine, with the warm-up excluded from the reported percentile. Across 15 MVTec categories and 6,000 images at batch size one with CUDA synchronization, mean/P95 latency was 2.3292/2.9298 ms for Fast, 3.6269/4.0860 ms for Full, and 5.1899/6.7302 ms for Risk (probe plus Full). Risk includes both local and full operations, so this measurement defines a boundary rather than an end-to-end deployment benchmark. The batch-throughput checks cover only three categories and remain an extension to the main analysis.

8 Reproducibility record

The reproducibility materials record the random seeds, category lists, budget values, derived tables, and figure source data. Code, environment specifications, and regeneration scripts are available in the project repository at https://github.com/cdhuangjin/fixed-budget-patch-routing.